

# EC601 — Control System

Semester VI • 3L:0T:0P • 3 Credits

## Module 1 Exam Notes

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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## 1. Introduction to Control Systems

A **control system** is an interconnection of physical components arranged to *regulate, direct, or command* the behaviour of another system so as to obtain a desired response. The fundamental goal is to **maintain a specified output** in spite of disturbances or parameter variations.

**Formal Definition:** A control system is a system in which the output quantity is controlled by varying the input quantity. It comprises a *controller*, a *plant* (process), and — in feedback systems — a *sensor* that measures the output.

### 1.1 Key Terminology

| Term                             | Definition                                                                                   |
|----------------------------------|----------------------------------------------------------------------------------------------|
| <b>Plant / Process</b>           | The physical object or process to be controlled (e.g. motor, furnace, tank).                 |
| <b>Controlled Variable</b>       | The output quantity measured and maintained at a desired level (e.g. speed, temperature).    |
| <b>Reference Input</b><br>$R(s)$ | The desired (set-point) value of the controlled variable, supplied externally.               |
| <b>Error Signal</b> $E(s)$       | $E = R - B$ ; difference between reference input and feedback signal. Drives the controller. |
| <b>Actuator</b>                  | Device (motor, valve, heater) that physically drives the plant based on controller output.   |
| <b>Sensor / Transducer</b>       | Measures the actual output and converts it into a form comparable to the reference signal.   |
| <b>Feedback Signal</b><br>$B(s)$ | Signal derived from the output (via sensor) fed back to the summing junction.                |
| <b>Disturbance</b>               | Unwanted external or internal signal that adversely affects the plant output.                |
| <b>Controller</b>                | Element that processes the error signal and generates corrective action $U(s)$ .             |

### 1.2 Industrial Control Examples

- Furnace Temperature Control:** Thermostat measures temperature, compares with set-point, switches heater on/off — classic *closed loop on-off* controller.
- DC Motor Speed Control:** Tachometer feeds back actual RPM; controller adjusts armature voltage to maintain desired speed despite load variations.
- Liquid Level Control:** Float/electronic sensor detects tank level; valve opened/closed to keep level at set value.

4. **Missile Guidance System:** Inertial/radar sensor tracks actual position and continuously corrects trajectory — high-speed closed loop system.
5. **Aircraft Auto-Pilot:** Gyroscopes and accelerometers measure altitude, heading, speed; auto-pilot adjusts control surfaces to match reference values.

## 2. Transfer Function

The transfer function is the primary mathematical tool to model, analyse, and design LTI control systems. It converts the time-domain differential equation into an algebraic expression in the  $s$ -domain (Laplace domain).

**Definition:** The **Transfer Function**  $G(s)$  of a linear time-invariant (LTI) system is the ratio of the Laplace transform of the *output* to the Laplace transform of the *input*, with all initial conditions assumed **zero**.

$$G(s) = \left. \frac{\mathcal{L}\{\text{output}\}}{\mathcal{L}\{\text{input}\}} \right|_{\text{zero I.C.}} = \frac{Y(s)}{U(s)}$$

### 2.1 Poles, Zeros, and Order

The denominator polynomial set to zero gives the **characteristic equation**. Its roots are the **poles** — they determine stability and transient response. Roots of the numerator are the **zeros**. The **order** is the highest power of  $s$  in the denominator.

### 2.2 Properties of Transfer Function

| Properties of Transfer Function |                                                                                                     |
|---------------------------------|-----------------------------------------------------------------------------------------------------|
| 1                               | Defined and valid only for <b>LTI</b> (Linear, Time-Invariant) systems.                             |
| 2                               | <b>Independent of input magnitude</b> — depends only on system parameters ( $R, C, L, K$ , etc.).   |
| 3                               | Setting denominator = 0 gives the <b>characteristic equation</b> ; its roots are the <b>poles</b> . |
| 4                               | Setting numerator = 0 gives the <b>zeros</b> of the system.                                         |
| 5                               | <b>Order</b> of system = degree of the denominator polynomial ( $n$ ).                              |
| 6                               | All initial conditions assumed <b>zero</b> — it is an <i>input-output</i> model only.               |
| 7                               | Does <b>not</b> describe internal (state) variables of the system.                                  |

### 2.3 Standard Form

$$G(s) = \frac{b_m s^m + b_{m-1} s^{m-1} + \dots + b_0}{a_n s^n + a_{n-1} s^{n-1} + \dots + a_0}, \quad n \geq m \text{ (proper system)}$$

## 2.4 Example — Series RC Circuit

**Given:** Series RC circuit. Input =  $V_i(s)$ , Output =  $V_o(s)$  (voltage across  $C$ ).  
Applying voltage divider in  $s$ -domain:

$$G(s) = \frac{V_o(s)}{V_i(s)} = \frac{1/sC}{R + 1/sC} = \frac{1}{1 + sRC} = \frac{1}{1 + s\tau}, \quad \tau = RC$$

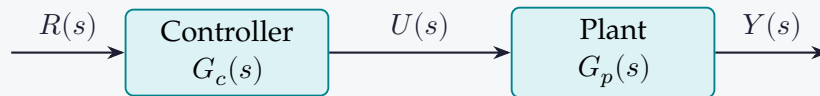
**Analysis:** First-order system ( $n = 1$ ). One pole at  $s = -1/\tau$ , no finite zeros. Time constant  $\tau = RC$  determines charging speed.

## 3. Open Loop Control System

In an open loop system, the controller generates a control signal based *only* on the reference input. The actual output is **never measured or compared** — the system cannot self-correct.

**Definition:** An **open loop control system** is one in which the control action is *independent of the output*. The output has no effect on the control signal.

### Open Loop Block Diagram



$$\frac{Y(s)}{R(s)} = G_c(s) \cdot G_p(s)$$

### 3.1 Characteristics of Open Loop Systems

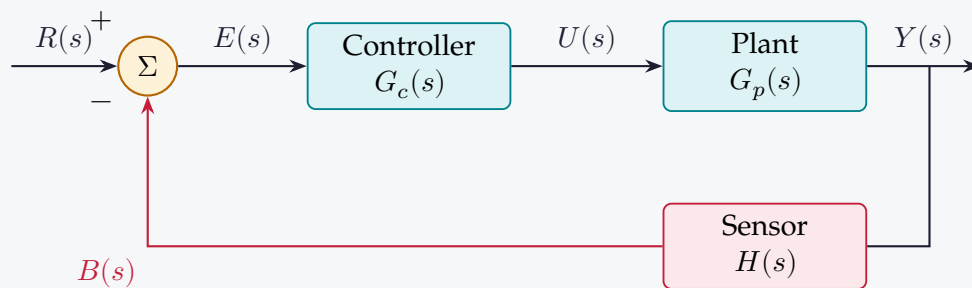
| Feature       | Open Loop Behaviour                                                                   |
|---------------|---------------------------------------------------------------------------------------|
| Accuracy      | Low — errors from disturbances or parameter changes are not corrected.                |
| Stability     | Generally stable; no feedback loop to introduce phase shift or oscillation.           |
| Complexity    | Simple design; fewer components; easy to maintain.                                    |
| Cost          | Lower — no sensors, feedback elements, or comparators needed.                         |
| Disturbance   | Output permanently affected; no self-correction mechanism.                            |
| Calibration   | Must be re-calibrated frequently to maintain accuracy.                                |
| Real Examples | Washing machine (timed cycle), traffic signal timer, electric toaster, stepper motor. |

## 4. Closed Loop (Feedback) Control System

A closed loop system **continuously monitors the output** and uses the measured value to correct the control action. The difference between reference and actual output is the **error**, which drives the controller.

**Definition:** A closed loop (feedback) control system is one in which the output is measured, compared with the reference input, and the resulting *error signal* is used to reduce the deviation. Feedback is typically **negative**.

### Closed Loop Block Diagram (Negative Feedback)



### 4.1 Closed Loop Transfer Function (CLTF)

**Derivation (Unity Feedback,  $H(s) = 1$ ):**

$$\begin{aligned} E(s) &= R(s) - Y(s) \\ Y(s) &= G(s) \cdot E(s) = G(s)[R(s) - Y(s)] \\ Y(s)[1 + G(s)] &= G(s) \cdot R(s) \end{aligned}$$

**Unity Feedback ( $H(s) = 1$ ):**

$$\frac{Y(s)}{R(s)} = \frac{G(s)}{1 + G(s)}$$

**Non-Unity Feedback ( $H(s) \neq 1$ ):**

$$\frac{Y(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}, \quad G(s) = G_c(s) \cdot G_p(s)$$

$G(s)H(s)$  = **open loop transfer function (OLTF)**.  $1 + G(s)H(s) = 0$  = **characteristic equation**.

## 4.2 Open Loop vs. Closed Loop — Comparison

| Parameter          | Open Loop                     | Closed Loop                         |
|--------------------|-------------------------------|-------------------------------------|
| Feedback           | Absent                        | Present (negative)                  |
| Accuracy           | Lower                         | Higher                              |
| Stability          | More stable                   | May become unstable                 |
| Disturbance effect | High; no correction           | Reduced; self-correcting            |
| Bandwidth          | Narrow                        | Wider                               |
| Sensitivity        | High to parameter changes     | Reduced (robust)                    |
| Cost & Complexity  | Low                           | Higher                              |
| Real Examples      | Toaster, timer, stepper motor | Thermostat, servo drive, auto-pilot |

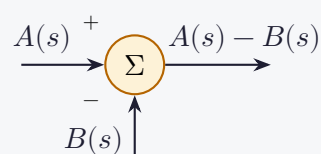
## 5. Block Diagram Reduction

A **block diagram** is a pictorial representation of a control system. For complex multi-loop systems, **algebraic reduction rules** simplify the diagram step-by-step until a single overall TF  $Y(s)/R(s)$  is obtained.

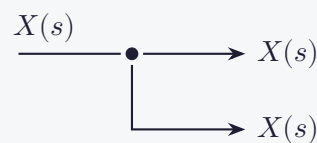
**Summing junction:** Circle with  $+/−$  signs; algebraically adds/subtracts incoming signals.

**Takeoff point:** Node from which the same signal is distributed to multiple branches unchanged.

### Summing Junction and Takeoff Point



Summing Junction



Takeoff Point

## 5.1 Reduction Rules

| Operation                   | Description                                              | Equivalent TF                  |
|-----------------------------|----------------------------------------------------------|--------------------------------|
| <b>Blocks in Series</b>     | Output of $G_1$ feeds input of $G_2$ ; same signal chain | $G_1(s) \cdot G_2(s)$          |
| <b>Blocks in Parallel</b>   | Same input, individual outputs summed at a junction      | $G_1(s) + G_2(s)$              |
| <b>Feedback Loop</b>        | Forward path $G$ , feedback path $H$ (negative feedback) | $\frac{G}{1 + GH}$             |
| <b>Summing pt. forward</b>  | Shift summing junction past block $G$ to the right       | Divide shifted branch by $G$   |
| <b>Summing pt. backward</b> | Shift summing junction past block $G$ to the left        | Multiply shifted branch by $G$ |
| <b>Takeoff pt. forward</b>  | Shift takeoff past block $G$ to the right                | Multiply branch by $G$         |
| <b>Takeoff pt. backward</b> | Shift takeoff past block $G$ to the left                 | Divide branch by $G$           |

## 5.2 Reduction Procedure (Step-by-Step)

- Step 1:** Identify and combine all **cascade (series)** blocks into a single block.
- Step 2:** Identify and combine all **parallel** paths into a single block.
- Step 3:** Eliminate all **feedback loops** using  $G/(1 \pm GH)$ .
- Step 4:** If needed, **shift** summing junctions or takeoff points to untangle crossed paths.
- Step 5:** Repeat steps 1–4 until only a **single block**  $Y(s)/R(s)$  remains.

### Exam Tip

Always maintain the correct sign (+/−) at every summing junction while shifting. Careless sign errors are the most common mistake in block diagram problems.

## 6. Signal Flow Graph (SFG) Analysis

A Signal Flow Graph is an alternative graphical method to represent and solve linear algebraic equations describing a control system. Unlike block diagrams — which require step-by-step reduction — an SFG is solved *directly* using **Mason's Gain Formula** in a single pass.

**Definition:** A **Signal Flow Graph (SFG)** is a directed graph of *nodes* (signals/variables) connected by *directed branches* (carrying gain/transmittance values). It is a compact representation of the system's simultaneous equations.

## 6.1 Basic Terminology

| Term                              | Meaning / Definition                                                                             |
|-----------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Node</b>                       | Represents a signal or variable in the system (shown as a small circle).                         |
| <b>Branch</b>                     | Directed arrow between two nodes; labelled with the <i>transmittance</i> (gain).                 |
| <b>Source Node</b>                | Input node — has <b>only outgoing</b> branches; no incoming branches.                            |
| <b>Sink Node</b>                  | Output node — has <b>only incoming</b> branches; no outgoing branches.                           |
| <b>Mixed Node</b>                 | A node with both incoming and outgoing branches (intermediate variable).                         |
| <b>Forward Path</b>               | Continuous directed path from source to sink in which <b>no node is visited more than once</b> . |
| <b>Forward Path Gain</b><br>$P_k$ | Product of all branch gains along the $k$ -th forward path.                                      |
| <b>Loop</b>                       | Closed directed path beginning and ending at the same node, <b>no node repeated</b> .            |
| <b>Loop Gain</b> $L_i$            | Product of all branch gains around the $i$ -th loop.                                             |
| <b>Non-Touching Loops</b>         | Two or more loops that do <b>not share any common node</b> .                                     |

## 6.2 Mason's Gain Formula

$$T = \frac{Y(s)}{R(s)} = \frac{\sum_{k=1}^N P_k \Delta_k}{\Delta}$$

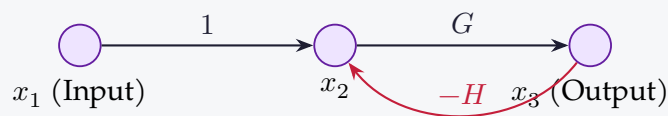
| Symbol             | Definition                                                                                                           |
|--------------------|----------------------------------------------------------------------------------------------------------------------|
| $N$                | Total number of forward paths from source to sink.                                                                   |
| $P_k$              | Gain of the $k$ -th forward path (product of branch gains along that path).                                          |
| $\Delta$           | Graph determinant = $1 - \sum_i L_i + \sum_{i,j} L_i L_j - \sum_{i,j,k} L_i L_j L_k + \dots$                         |
| $\sum L_i$         | Sum of gains of <b>all individual loops</b> in the graph.                                                            |
| $\sum L_i L_j$     | Sum of gain-products of all <b>pairs of non-touching loops</b> .                                                     |
| $\sum L_i L_j L_k$ | Sum of gain-products of all <b>triplets of mutually non-touching loops</b> .                                         |
| $\Delta_k$         | <b>Cofactor</b> of the $k$ -th path: value of $\Delta$ after removing all loops that <i>touch</i> forward path $k$ . |

### 6.3 Steps to Apply Mason's Formula

- 1. Identify forward paths:** Trace every path from source to sink (no repeated nodes). Note gain  $P_k$  of each.
- 2. Identify all loops:** Find every closed path. Compute loop gain  $L_i$  for each.
- 3. Find non-touching loop combinations:** Identify all pairs, triplets, etc. sharing no common node.
- 4. Compute graph determinant:**  $\Delta = 1 - \sum L_i + \sum L_i L_j - \sum L_i L_j L_k + \dots$
- 5. Compute cofactor  $\Delta_k$ :** For each forward path  $k$ , remove all loops touching that path from  $\Delta$ .
- 6. Apply Mason's formula:**  $T = \frac{\sum P_k \Delta_k}{\Delta}$ .

### 6.4 Worked Example — Simple Feedback System

#### SFG Representation



**Applying Mason's formula:**

- Forward path:  $P_1 = 1 \times G = G$
- Loop:  $L_1 = G \times (-H) = -GH$
- No non-touching loop pairs (only one loop).
- $\Delta = 1 - L_1 = 1 - (-GH) = 1 + GH$
- $\Delta_1 = 1$  (loop  $L_1$  touches forward path  $P_1$ , so removed from  $\Delta$ )

$$T = \frac{P_1 \Delta_1}{\Delta} = \frac{G \cdot 1}{1 + GH} = \frac{G}{1 + GH} \quad \checkmark$$

Matches the CLTF from block diagram reduction — both methods are equivalent.

### 6.5 SFG vs. Block Diagram — Comparison

| Aspect           | Block Diagram                                 | Signal Flow Graph                           |
|------------------|-----------------------------------------------|---------------------------------------------|
| Representation   | Blocks, summing junctions, takeoff points     | Directed nodes and weighted branches        |
| Reduction Method | Step-by-step algebra; multiple passes         | Mason's formula — single direct computation |
| Complex Systems  | Tedious; can become cluttered and error-prone | Compact, systematic, less error-prone       |
| Gain Formula     | No direct formula; reduce iteratively         | $T = \sum P_k \Delta_k / \Delta$            |
| Preferred For    | Simple systems; visual understanding          | Multi-loop systems; quick TF computation    |

## Quick Revision — Module 1

Control System | EC601

### Key Formulas at a Glance

|                                |                                                |
|--------------------------------|------------------------------------------------|
| <b>Transfer Function</b>       | $G(s) = Y(s)/U(s)$ (zero initial conditions)   |
| <b>CLTF (Unity feedback)</b>   | $T(s) = G(s)/[1 + G(s)]$                       |
| <b>CLTF (Non-unity)</b>        | $T(s) = G(s)/[1 + G(s)H(s)]$                   |
| <b>Characteristic Equation</b> | $1 + G(s)H(s) = 0$                             |
| <b>Series blocks</b>           | $G_{eq} = G_1 \cdot G_2$                       |
| <b>Parallel blocks</b>         | $G_{eq} = G_1 + G_2$                           |
| <b>Feedback reduction</b>      | $G_{eq} = G/(1 \pm GH)$                        |
| <b>Mason's Gain Formula</b>    | $T = \frac{\sum_k P_k \Delta_k}{\Delta}$       |
| <b>Graph Determinant</b>       | $\Delta = 1 - \sum L_i + \sum L_i L_j - \dots$ |

### One-Line Definitions

- **Control System:** Interconnection of components to regulate output to a desired value.
- **Transfer Function:** Ratio of Laplace output to Laplace input at zero initial conditions.
- **Poles:** Roots of denominator of  $G(s)$  — determine stability and transient response.
- **Zeros:** Roots of numerator of  $G(s)$  — affect frequency response shape.
- **Open Loop:** Control action independent of output; no self-correction.
- **Closed Loop:** Output fed back and compared with reference; error drives controller.
- **Error Signal:**  $E(s) = R(s) - B(s)$ ; difference between reference and feedback.
- **SFG:** Directed graph of nodes and branches representing system equations.
- **Forward Path:** Path from source to sink with no repeated nodes.
- **Loop Gain:** Product of all branch gains around a closed loop.
- **Cofactor  $\Delta_k$ :** Graph determinant  $\Delta$  with all loops touching path  $k$  removed.

### Frequently Asked / Viva Points

#### Q1. Why are initial conditions zero in TF?

TF is an input-output model; it characterises the system, not a specific response.

## Frequently Asked / Viva Questions (contd.)

Zero I.C. ensures the TF is a property of the system alone.

**Q2. What is the order of a system?**

Highest power of  $s$  in the denominator of  $G(s)$ .

**Q3. Difference between open loop and closed loop?**

Open loop: no feedback, no self-correction. Closed loop: feedback present, error-driven, self-correcting.

**Q4. What is the characteristic equation?**

$1 + G(s)H(s) = 0$ . Its roots (poles of CLTF) determine stability.

**Q5. When is a system proper?**

When degree of denominator  $\geq$  degree of numerator ( $n \geq m$ ).

**Q6. What is a takeoff point?**

A node from which the same signal branches to multiple paths without change.

**Q7. What is  $\Delta_k$  in Mason's formula?**

Cofactor of the  $k$ -th forward path —  $\Delta$  computed after removing all loops that touch path  $k$ .

**Q8. Can SFG and block diagram give different results?**

No — both are equivalent representations; Mason's formula gives the same TF as block diagram reduction.

**Q9. What are non-touching loops?**

Loops that share no common node. Their gain products appear in  $\Delta$ .

**Q10. Why is negative feedback preferred?**

It reduces error, improves accuracy, reduces sensitivity to parameter variations, and widens bandwidth.

## Comparison Snapshot

| Property    | Open Loop     | Closed Loop    |
|-------------|---------------|----------------|
| Feedback    | No            | Yes (negative) |
| Accuracy    | Low           | High           |
| Stability   | More stable   | Can oscillate  |
| Cost        | Low           | High           |
| Disturbance | Not corrected | Self-corrected |